

CA1 – DFAs, NFAs, and e-NFAs

1



In order to diminish the size and to speed-up the processing of ϵ -NFAs (especially the ones obtained by the scheme to convert regular expressions to ϵ -NFAs), a team intends to remove ϵ transitions from the ϵ -NFAs.

The team suggested the following two-step (A and B, and with first application of A and then of B) scheme to remove the ϵ -transitions:

A. While there are modifications in the FA:

- For each pair of states (q_i, q_j) with a single transition, an ϵ -transition, from q_i to q_j , only a single input transition to q_i and a single output transition from q_j , merge the two states in one state q_i-q_j (the transition from q_j becomes transition from q_i-q_j ; and the transition to q_i becomes transition to q_i-q_j).

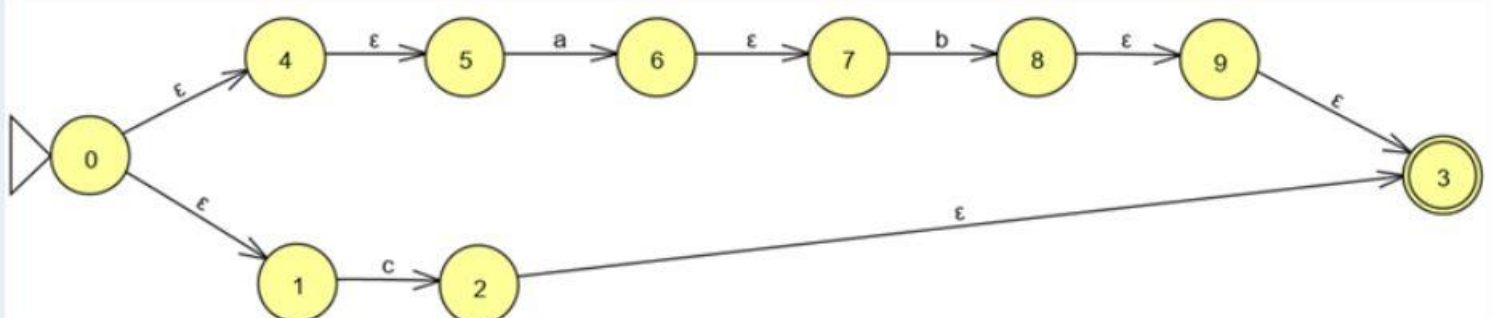
B. While there are modifications in the FA:

- for each $\delta(q_i, \epsilon) = \{q_j\}$ and $q_i \neq q_j$, merge q_i-q_j (q_i becomes q_i-q_j and the transitions from q_i and from q_j become transitions from q_i-q_j ; and the transitions to q_i and q_j become transitions to q_i-q_j , then q_j and its transitions are removed from the FA in the case of the only transition from q_i and q_j is the ϵ -transition).

For both steps A and B, if q_i is the initial state then q_i-q_j becomes the initial state. q_i-q_j becomes a final state if q_i or q_j are final states.

2

Consider the following input e-NFA (obtained from the regular expression $c+ab$):



3

Apply the A-step of the scheme to the e-NFA and show the resultant FA. 
(1 Ponto)

4

Is the new FA equivalent to the e-NFA? 
(1 Ponto)

☐ YES☐ NO

5

Apply the B-step of the scheme to the FA resultant from the A-step and show the new FA. 
(1 Ponto)

6

Is the new FA equivalent to the original e-NFA? 
(1 Ponto)

☐ TRUE☐ FALSE

7

Is proposed scheme valid for any input e-NFA? 
(1 Ponto)

☐ TRUE☐ FALSE

8

The team, instead of proving the validity of the transformation for any e-NFA, decided, each time the transformation is used, to verify the equivalence between the two FAs (the input and the output ones) by using a program specially developed for that. Suggest the main steps involved in the verification process of that program.
(3 Pontos)